



Spring MVC with Boot

- ♦ You can develop Spring MVC application with Spring Boot in two ways.
 - Develop as FAT JAR file.
 - Develop as WAR file
- ♦ FAT Jar is self executable application which contains Application Code + Embedded Container
- ♦ You can execute the FAT jar as follows.
 - Java -jar HelloApp.jar
- ♦ WAR file is deployable application which can be deployed in any External Container to run.

demo1-mvc-boot : Spring MVC with Boot (with JSP)

Working Steps:

A) Setup the Project

- 1) Create the maven Project.
 - a) Open the Eclipse required Workspace
 - b) Select New -> Maven Project.
 - c) Check the checkbox for create Simple Project(skip the Archetype selection) and Click on Next Button.
 - d) Provide the following
 - Group Id: com.coursecube.springboot
 - Artifact Id : **demo1-mvc-boot**
 - Version : 1.0
 - packaging : jar
 - Name : **demo1-mvc-boot**and Click On finish button.
- 2) Remove the following two Source Folders.
 - a. src/test/java
 - b. src/test/resources
- 3) Create application.properties under src/main/resources
- 4) Update application.properties as follows
server.port=12345
- 5) Create the Source Folder called src/main/webapp
- 6) Create the Folder called WEB-INF under src/main/webapp
- 7) Create the Folder called myjsps under WEB-INF



8) Update pom.xml by adding the following dependencies.

```
<parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>2.1.16.RELEASE</version>
</parent>

<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
        <groupId>org.apache.tomcat.embed</groupId>
        <artifactId>tomcat-embed-jasper</artifactId>
    </dependency>

    <dependency>
        <groupId>jstl</groupId>
        <artifactId>jstl</artifactId>
        <version>1.2</version>
    </dependency>

    <dependency>
        <groupId>taglibs</groupId>
        <artifactId>standard</artifactId>
        <version>1.1.2</version>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-devtools</artifactId>
    </dependency>
</dependencies>
```

9) You can see the Error in Maven Project.

Fix the Error by updating the Maven Project.

Select the Project , right click and then select Maven -> Update Project.

10) Create the Package called com.coursecube.springboot under src/main/java



11) Write Bean Configuration class – JLCWebConfig.java

12) Configure View Resolver in JLCWebConfig.java

```
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{

    @Bean
    public ViewResolver viewResolver() {
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();
        resolver.setPrefix("/WEB-INF/myjsps/");
        resolver.setSuffix(".jsp");
        resolver.setViewClass(JstlView.class);
        return resolver;
    }
}
```

13) Write Boot Main Class – StartMyBootApplication.java

```
@SpringBootApplication
public class StartMyBootApplication {
    public static void main(String[] args) {
        SpringApplication.run(StartMyBootApplication.class, args);
    }
}
```

14) Run the Boot Application

You will see – Whitelabel Error Page means DS is Ready.

B) Develop the Usecase : show index page

http://localhost:12345/ => You are sending the request for /

15) Write index.jsp under WEB-INF/myjsps folder.

```
<!DOCTYPE html>
<html>
<body>
<h1 class="text-center"> Welcome to CourseCube !!!</h1>
</body>
</html>
```



16) Write IndexController.java

```
@Controller
public class IndexController {
    @GetMapping("/")
    public String showIndexPage() {
        System.out.println("IndexController - showIndexPage()");
        return "index";
    }
}
```

17) Run the Boot Application

18) Open the browser and hit <http://localhost:12345/>
If every thing fine you will see index.jsp

D)Develop the Usecase : show books

<http://localhost:12345/showBooks>

19) Add hyperlink in index.jsp

```
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>
```

20) Write Book.java under com.coursecube.springboot

21) Write BooksController.java under com.coursecube.springboot

22) Write booksList.jsp under WEB-INF/myjsps

23) Run the Boot Application and Test

Open the browser and hit <http://localhost:12345/>

Lab17: Files Required

1. index.jsp	2. booksList.jsp
3. Book.java	4. IndexController.java
5. BooksController.java	6. JLCWebConfig.java
7. StartMyBootApplication.java	8. application.properties
9. pom.xml	



Welcome to CourseCube !!!

Latest Books from Srinivas Dande

Book Id	Book Name	Author	Price	Publications		
101	Master Spring Boot	Srinivas Dande	10000.0	JLC	View Book	Edit Book
102	Master MicroServices	Srinivas Dande	10000.0	JLC	View Book	Edit Book
103	Master Angular	Srinivas Dande	10000.0	JLC	View Book	Edit Book
104	Master React	Srinivas Dande	10000.0	JLC	View Book	Edit Book
105	Master Java Full Stack	Srinivas Dande	10000.0	JLC	View Book	Edit Book

1. index.jsp

```
<!DOCTYPE html>
<html>
<head>
  <title>JLC Bookstore</title>
</head>

<body>
<h1 class="text-center"> Welcome to CourseCube !!!</h1>
<br/>
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>

</body>
</html>
```

2. booksList.jsp

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<html> <head>
<title>JLC Bookstore</title>
</head>
<body>
<div class="container">
<h1 class="text-center"> Welcome to CourseCube !!!</h1>
<div class="alert alert-primary" role="alert">
<h2 class="text-center"> Latest Books from Srinivas Dande </h2>
</div>
<br/>
```



```
<table class="table table-striped table-bordered table-hover" style="font-size:20px;">
<thead class="thead-light ">
<tr> <th> Book Id </th>
<th> Book Name </th>
<th> Author</th>
<th> Price </th>
<th> Publications</th>
<th> </th>
<th> </th> </tr>
</thead>
<tbody>
<c:forEach var="mybook" items="${MyBooks}">
<tr>
<td> <a href="#"> ${mybook.bid} </a> </td>
<td> ${mybook.bname} </td>
<td> ${mybook.author} </td>
<td> ${mybook.price} </td>
<td> ${mybook.pub}</td>
<td> <button type="button" class="btn btn-success btn-lg">View Book</button> </td>
<td> <button type="button" class="btn btn-primary btn-lg">Edit Book</button> </td>
</tr>
</c:forEach>
</tbody>
</table>
</div>
</body>
</html>
```

3. Book.java

```
spackage com.coursecube.springboot;
```

```
import java.math.BigDecimal;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
public class Book {
    private Integer bid;
    private String bname;
    private String author;
    private BigDecimal price;
    private String pub;
```



```
public Book() {}  
//Constrcutors  
//Setters and Getters  
}
```

4. **IndexController.java**

```
package com.coursecube.springboot;  
  
import org.springframework.stereotype.Controller;  
import org.springframework.web.bind.annotation.GetMapping;  
/*  
 * @Author : Srinivas Dande  
 * @Company : CourseCube Private Limited  
 * @Website : www.coursecube.com  
 */  
@Controller  
public class IndexController {  
  
    @GetMapping("/")  
    public String showIndexPage() {  
        System.out.println("IndexController - showIndexPage()");  
        return "index";  
    }  
}
```

5. **BooksController.java**

```
package com.coursecube.springboot;  
  
import java.math.BigDecimal;  
import java.util.*;  
import javax.servlet.http.HttpSession;  
import org.springframework.stereotype.Controller;  
import org.springframework.web.bind.annotation.GetMapping;  
/*  
 * @Author : Srinivas Dande  
 * @Company : CourseCube Private Limited  
 * @Website : www.coursecube.com  
 */  
@Controller  
public class BooksController {
```



```
@GetMapping("/showBooks")
public String getBooks(HttpSession session) {
    System.out.println(" BooksController - getBooks()");

    List<Book> blist=new ArrayList<>();
    Book mybook1=new Book(101,"Master Spring Boot","Srinivas Dande",
    BigDecimal.valueOf(10000.0),"JLC");

    Book mybook2=new Book(102,"Master MicroServices","Srinivas Dande",
    BigDecimal.valueOf(10000.0),"JLC");
    Book mybook3=new Book(103,"Master Angular","Srinivas Dande",
    BigDecimal.valueOf(10000.0),"JLC");

    Book mybook4=new Book(104,"Master React ","Srinivas Dande",
    BigDecimal.valueOf(10000.0),"JLC");

    Book mybook5=new Book(105,"Master Java Full Stack","Srinivas Dande",
    BigDecimal.valueOf(10000.0),"JLC");

    blist.add(mybook1); blist.add(mybook2); blist.add(mybook3);
    blist.add(mybook4); blist.add(mybook5);

    session.setAttribute("MyBooks", blist);

    return "booksList";
}
}
```

6. JLCWebConfig.java

```
package com.coursecube.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.web.servlet.ViewResolver;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.view.InternalResourceViewResolver;
import org.springframework.web.servlet.view.JstlView;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */
@SpringBootApplication
```



```
public class JLCWebConfig implements WebMvcConfigurer{
    @Bean
    public ViewResolver viewResolver() {
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();
        resolver.setPrefix("/WEB-INF/myjsps/");
        resolver.setSuffix(".jsp");
        resolver.setViewClass(JstlView.class);
        return resolver;
    }
}
```

7. StartMyBootApplication.java

```
package com.coursecube.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.web.servlet.support.SpringBootServletInitializer;
/*
 * @Author : Srinivas Dande
 * @Company : CourseCube Private Limited
 * @Website : www.coursecube.com
 */

@SpringBootApplication
public class StartMyBootApplication {
    public static void main(String[] args) {
        SpringApplication.run(StartMyBootApplication.class, args);
    }
}
```

8. application.properties

```
server.port=12345
```

9. pom.xml

```
<project...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.coursecube.springboot</groupId>
<artifactId>demo1-mvc-boot</artifactId>
<version>1.0</version>
<name>demo1-mvc-boot</name>
<url>https://www.coursecube.com</url>

<parent>
```



```
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-starter-parent</artifactId>  
<version>2.1.16.RELEASE</version>  
</parent>
```

```
<dependencies>  
<dependency>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-starter-web</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>org.apache.tomcat.embed</groupId>  
<artifactId>tomcat-embed-jasper</artifactId>  
</dependency>
```

```
<dependency>  
<groupId>jstl</groupId>  
<artifactId>jstl</artifactId>  
<version>1.2</version>  
</dependency>  
<dependency>  
<groupId>>taglibs</groupId>  
<artifactId>standard</artifactId>  
<version>1.1.2</version>  
</dependency>
```

```
<dependency>  
<groupId>org.webjars</groupId>  
<artifactId>bootstrap</artifactId>  
<version>4.3.1</version>  
</dependency>
```

```
<dependency>  
<groupId>org.webjars</groupId>  
<artifactId>jquery</artifactId>  
<version>3.4.1</version>  
</dependency>
```

```
<dependency>  
<groupId>org.springframework.boot</groupId>
```



```
<artifactId>spring-boot-devtools</artifactId>  
</dependency>  
</dependencies>
```

```
<build>  
<finalName>MyLab17</finalName>  
<plugins>
```

```
<plugin>  
<groupId>org.springframework.boot</groupId>  
<artifactId>spring-boot-maven-plugin</artifactId>  
</plugin>  
</plugins>  
</build>
```

```
</project>
```